



Myntra Product & Database Dissection

1. Company overview

Myntra is India's leading fashion and lifestyle e-commerce platform, founded in 2007 and acquired by Flipkart in 2014.

It connects **millions of users, thousands of brands** and **hundreds of sellers**, delivering a **personalized, seamless shopping experience**.

The platform offers products across multiple categories including:

- Clothing
- Footwear
- Accessories
- Beauty Products

Key Technologies:

- Data Analytics & Recommendation Systems-

Personalized product suggestions.

- **Optimized Database Architecture** - handle millions of concurrent transactions
- **Scalable Cloud infrastructure** - high availability & low latency

The **database is the backbone**, enabling inventory management, order processing, payments, and customer engagement.

2. Real-World Problems & Solutions

Problem	Real-World Challenge	Myntra Solution	Business Impact
Limited Access to Brands	Customers in smaller cities cannot access multiple brand easily	Online catalog with thousands of brands filters of size , color, price, category	Expands reach,increase sales,enhance customer choice
Difficulty Comparing Product	Physical stores limit simultaneous comparison	High-resolution images,detailed descriptions, rating & reviews	Reduce purchase hisitation,increase conversion
Lack of Trust in Online	Customers hesitant to buy unseen products	Review & rating system providing social proof	Build confidence, reduce returns,improves repeat purchases
Order Tracking and Management	Customers want transparency in delivery	Order history, real time shipment tracking , notifications	Improves transparency ,reduces support tickets,

			enhances loyalty
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3. Core Platform Features

Feature	Function	Product/Business Insight
User Accounts	Stores persona details,address, purchase history, wishlist	Supports retention, personalized campaigns and engagement analytics
Product Catalog	Hierarchical categories,brand association, multiple images	Enables data-driven merchandising, demand forecasting , product analytics
Shopping Cart	Tracks intended purchases, Calculates pricing, applies promotions	Identifies intent , abandoned carts, upsell/cross-sell Opportunities
Secure Payments	Multiple Payment Methods, real-time logging	Ensures trust, regulatory, compliance, fraud detection
Reviews Ratings	Captures post- purchase feedback	Improves recommendations,quality control, product strategy
Whishlist	Tracks products users are interested in	Drives personlized marketing campaigns and future conversons
Inventory	Real-time stock updates	Supports dynamic pricing , automated replenishment, stock forecasting

4. SQL & Analytics Focus

Core SQL concepts Applied:

- Relational integrity with **primary & foreign keys**
- **Many-to-many relationships** via junction tables (OrderItems, CartItems)
- Real-world analytics enabled via structured tables (sales, inventory, user behavior)

Sample Queries():

- Top -selling products last month

```
SELECT p.product_name , SUM(oi.quantity) as total_sold
FROM order_items oi
JOIN products p ON oi.product_id = p.product_id
JOIN orders o ON oi.order_id = o.order_id
WHERE o.order_date >= '2026-02-01'
GROUP BY p.product_name
ORDER BY total_sold DESC
Limit 10;
```

-- Users with abandoned carts

```
SELECT u.user_id, u.name, COUNT(ci.cart_item_id) AS
items_in_cart
FROM users u
JOIN cart c ON u.user_id = c.user_id
JOIN cart_items ci ON c.cart_id = ci.cart_id
WHERE c.created_at >= NOW() - INTERVAL '7 days'
GROUP BY u.user_id, u.name
HAVING COUNT(ci.cart_item_id) > 0;
```

-- Average rating per brand

```
SELECT b.brand_name, AVG(r.rating) AS avg_rating
FROM brands b
JOIN products p ON b.brand_id = p.brand_id
JOIN reviews r ON p.product_id = r.product_id
GROUP BY b.brand_name
ORDER BY avg_rating DESC;
```

5. Key Entities & Attributes

Entity	Key Attributes	Product/ Business Insight
Users	UserID, Name, Email,Phone, RegistrationDate	Segment users for retention & marketing
UserAddresses	AddressID, UserID, AddressLine , City, State,Pincode	Enables location-based offers & optimized delivery
Brands	BrandID, BrandName	Tracks brand merchandising strategy
Categories	CategoryID,CategoryName, ParentCategoryID	Supports catalog navigation ^ and analytics
Products	ProductID, ProductName, BrandID, CategoryID,Price, Description	Core entity for revenue, analytics, product strategy
Products Images	ImageID, ProductID, ImageURL, IsPrimary	Enhances UI/UX , visual merchandising A/B testing
Inventory	InventoryID, ProductID, StockQuantity, WarehouseLocation	Stock forecasting, automated replenishment
Cart & Cartitems	CartID, CartItemID, ProductID, Quantity	Tracks purchase intent, abandoned cart analytics
Orders& OrderItems	OrderID, ProductID, Quantity, Price	Revenue & fulfillment analytics
Payments	PaymentID, OrderID, PaymentsMethod, PayementsStatus	Financial reporting fraud detection
Shipments	ShipmentID, OrderID, CourierName, TrackingNumber status	Logistic efficiency, SLA monitoring
Reviews	ReviewID, ProductID, UserID, Rating, Comment	Customer satisfaction, product improvement, recommendation data

Whishlist	WhishlistID, UserID, ProductID	Tracks interest, future conversion, marketing campaigns
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● 6.Relationships & ER Logic

- **One-to-Many:** Users -> Orders, Orders->OrderItems,
- Products-> Inventory
- **Many-to-Many :** Orders-> Products (OrderItems),
- Cart<->Products(CartItems)
- **Hierarchical Modeling :** Categories-> Subcategories
- **Data Integrity :** Foreign key ensure consisten, reliable data

.7. Industry & Product Perspective

Business Impact of SQL & Database Knowledge:

- **Revenue Optimization:** Identify top products, buying patterns and sales trends.
- **Inventory Forecasting :** Real-time stock management,avoid stockouts
- **User Retention & Engagement :** Abandoned cart alerts, wishlist analysis, review analytics.
- **Product Strategy:** Brand performance , category insights ,price elasticity.
- **Operational Reliability:** Strong schema supports analytics ,dashboards and ML pipelines

8. Conclusion -

This project demonstrates how a robust SQL-backed database enables a large-scale e-commerce platform like Myntra to operate efficiently and scale.

Key Takeways:

Well-structured entities & relationship support product management , order fulfillment , inventory

Control and payments.

Data-driven insights guide marketing, merchandising, and product strategy.

Real-world SQL queries i

